



Overview of SAP Cloud Application Programming Model

LT104

PUBLIC

Speakers

Las Vegas

September 24–27, 2019

Christian Georgi

Daniel Hutzel

Barcelona

October 8-10, 2019

Anne Keller

Adrian Goerler

Bangalore

November 13-15, 2019

Gottipati Balakrishna



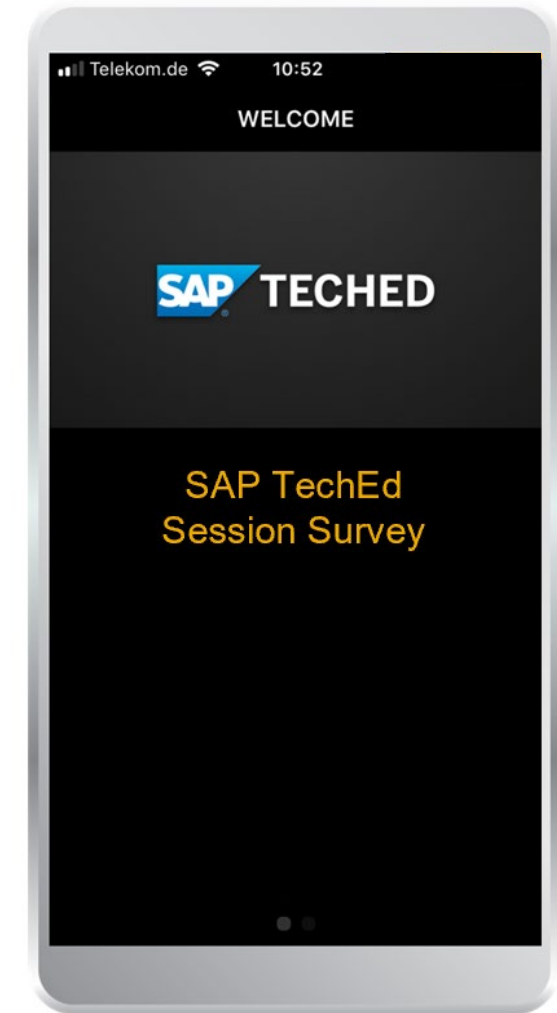
Take the **session survey**.

We want to hear from you!

Complete the session evaluation for this session **LT104** on the SAP TechEd mobile app.



Download the app from
iPhone App Store or Google Play.



Disclaimer

The information in this presentation is confidential and proprietary to SAP and may not be disclosed without the permission of SAP. Except for your obligation to protect confidential information, this presentation is not subject to your license agreement or any other service or subscription agreement with SAP. SAP has no obligation to pursue any course of business outlined in this presentation or any related document, or to develop or release any functionality mentioned therein.

This presentation, or any related document and SAP's strategy and possible future developments, products and or platforms directions and functionality are all subject to change and may be changed by SAP at any time for any reason without notice. The information in this presentation is not a commitment, promise or legal obligation to deliver any material, code or functionality. This presentation is provided without a warranty of any kind, either express or implied, including but not limited to, the implied warranties of merchantability, fitness for a particular purpose, or non-infringement. This presentation is for informational purposes and may not be incorporated into a contract. SAP assumes no responsibility for errors or omissions in this presentation, except if such damages were caused by SAP's intentional or gross negligence.

All forward-looking statements are subject to various risks and uncertainties that could cause actual results to differ materially from expectations. Readers are cautioned not to place undue reliance on these forward-looking statements, which speak only as of their dates, and they should not be relied upon in making purchasing decisions.

SAP Cloud Application Programming Model

An introduction

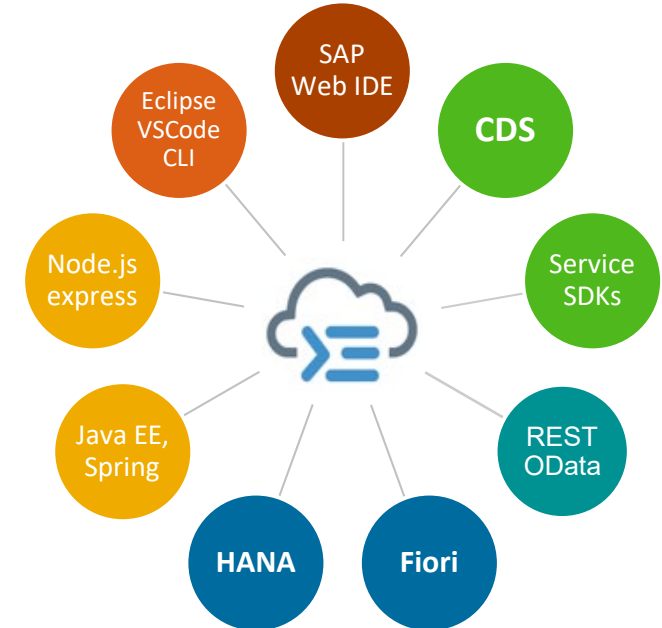


Introduction

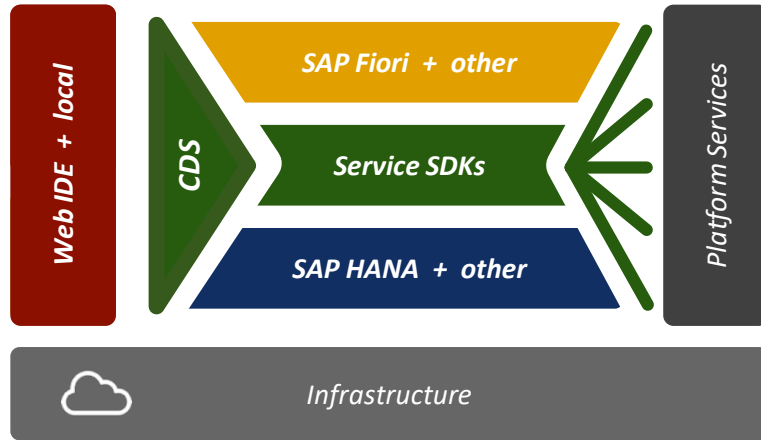
*The SAP Cloud Application Programming Model is an **opinionated**, yet **open** framework of tools, languages and libraries to efficiently build enterprise-grade services and applications.*

*It guides developers along a 'golden path' of proven **best practices**, while minimizing boilerplate so they can **focus on their domain** problems at hand.*

*The framework features a mix of broadly adopted **open-source** and **SAP** tools and technologies.*



Introduction

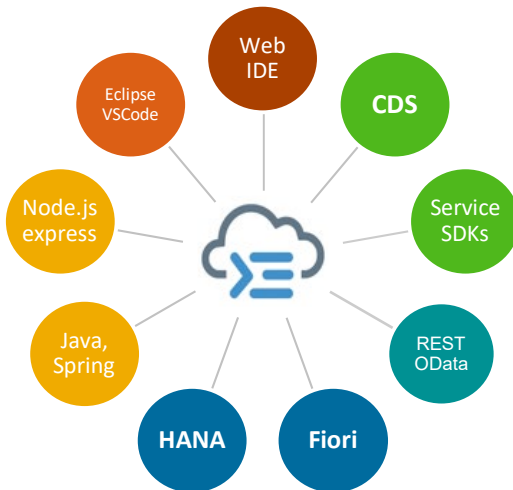


We complement *cloud-native technologies* with...

- a **CDS-based Services** framework...
- w/ intrinsic **enterprise & cloud** qualities

w/ *1st-class support* by and for...

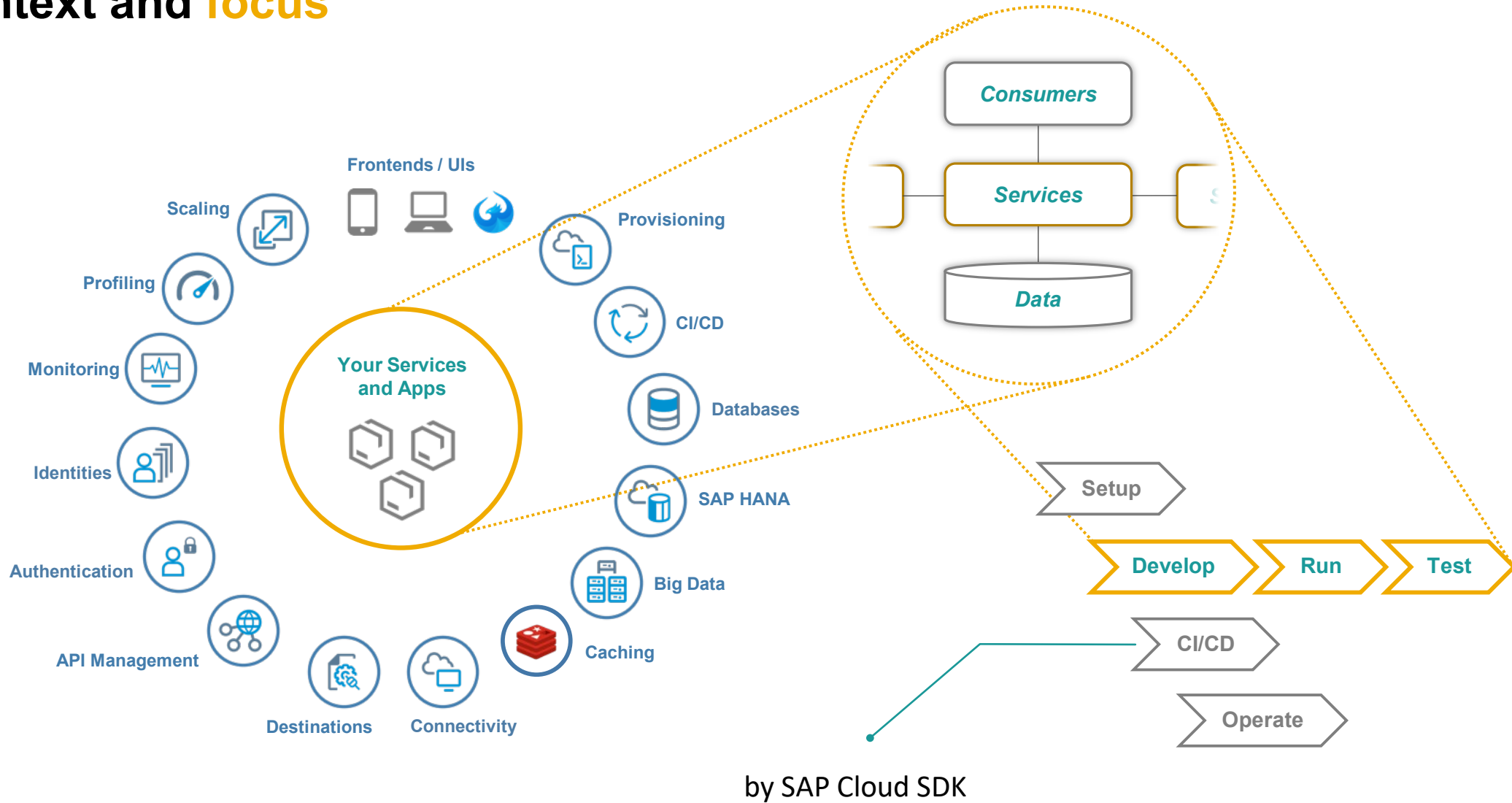
- **SAP Web IDE**
 - **SAP Fiori**
 - **SAP HANA**
 - **S/4 Extension** scenarios, ...
- } *without lock-in!*



SAP S/4HANA®
planned: **SAP SuccessFactors** 
SAP Concur 
SAP Fieldglass 

- **Guidance** and **best practices** leveraging proven SAP technologies + Open Source software
- A *consistent* end-to-end **programming model** for *enterprise* services & applications

Context and focus



Focus topics

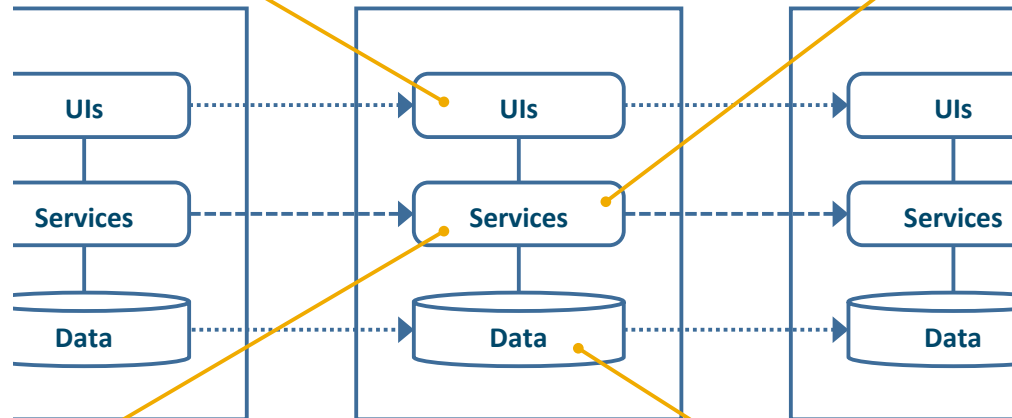


Serving Fiori UIs

- *OData v2 + v4*
- *Draft* support
- *\$batch* support, ...

Providing Services

- Service Definitions in CDS
- Service Providers
- Event Handlers → generic + custom
- **Emitting Events**



Consuming Services

- Import Service Definitions from OData, ...
- Mash-ups with local services
- Service Client APIs
- Connectivity & Resilience by SDKs
- **Subscribing to & Handling Events**

Database Support

- Out-of-the-box: *HANA*, *sqlite*, next: *PostgreSQL*
- Dynamic Data Access layers (*cds.sql*)
- Schema evolution

Demo

Best Practises and Benefits



Best Practices & Benefits...

Capture Intent on App Domain Level

» *What, not How.* «

```
using { Country, managed }  
from '@sap/cds/common';  
  
service ProductService {  
  
    entity Products : managed {  
        key ID : UUID;  
        title : localized String;  
        descr : localized String;  
        country : Country;  
    }  
}
```

Minimizing Boilerplate

Capture Intent on App Domain Level

```
using { Country, managed }  
from '@sap/cds/common';  
  
service ProductService {  
  entity Products : managed {  
    key ID : UUID;  
    title : localized String;  
    descr : localized String;  
    country : Country;  
  }  
}
```

» *What, not How.* «

It's not about less to type...

It's much more about:

- Conciseness and comprehensiveness
- Refocusing on your domain
- Detecting flaws early
- Single points to fix & optimize

i.e. about improving overall **quality**,
robustness and **adaptability**.

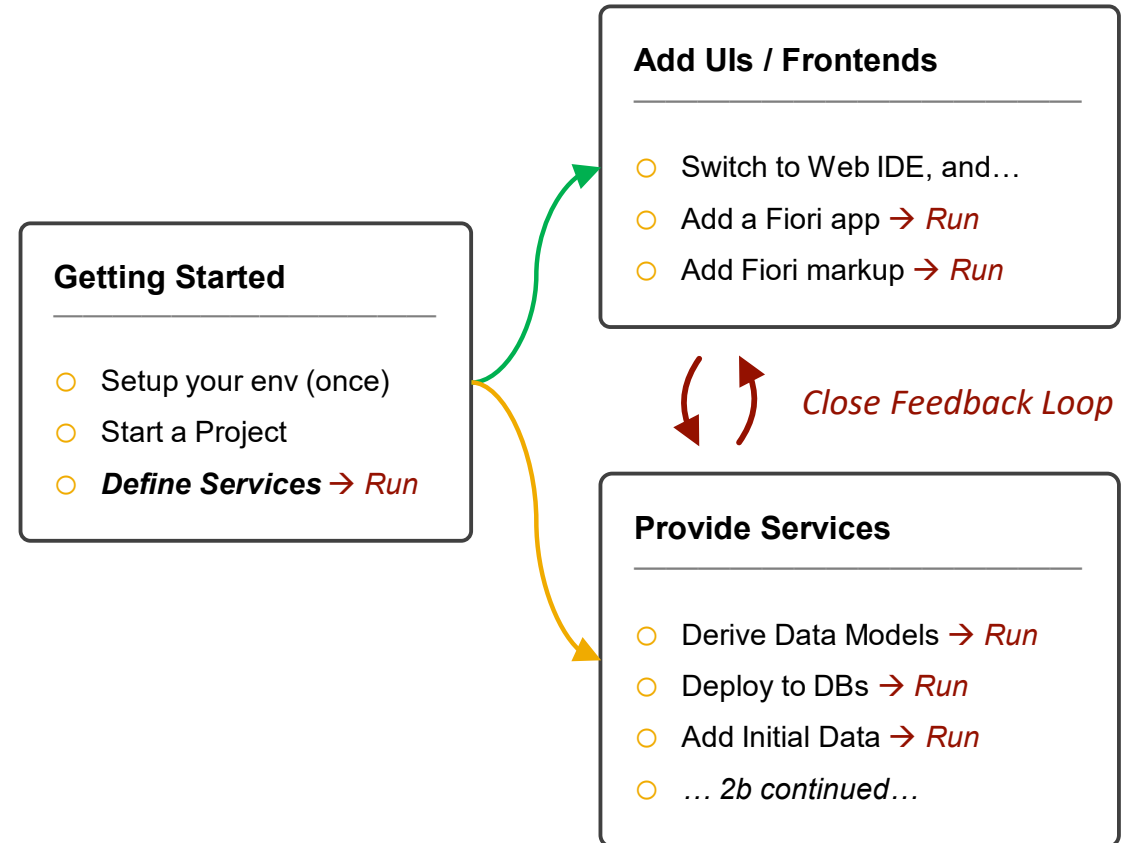
Best Practices & Benefits...

Capture Intent on App Domain Level

Accelerating Project Development

» *Run early* «

» *Grow as you go* «



- Start with services top-down
- Spin-off parallel workstreams (→ frontend / backend)
- Using the Service Definitions as their contracts

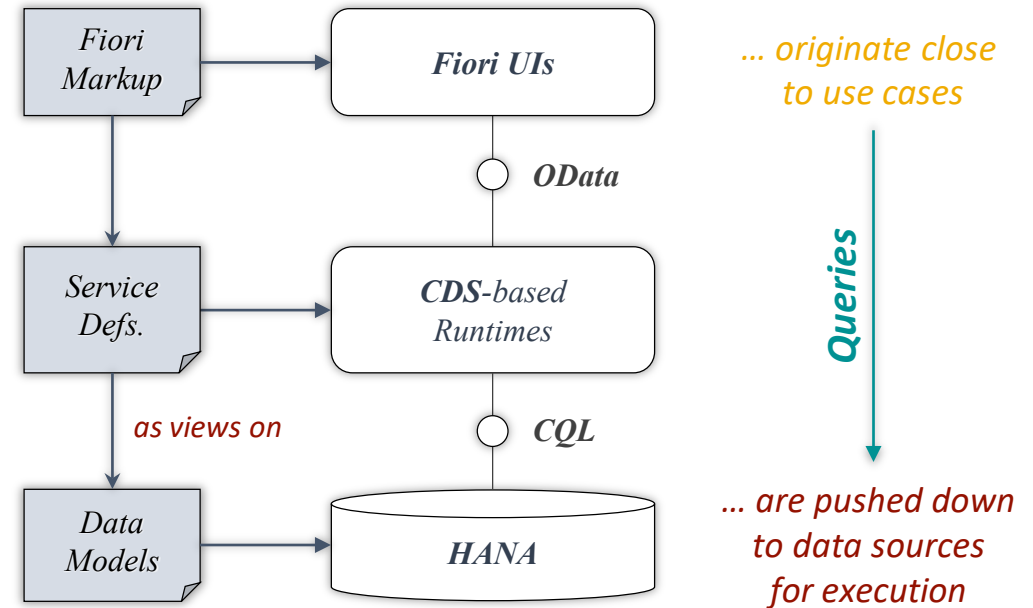
Best Practices & Benefits...

Capture Intent on App Domain Level

Accelerating Project Development

Leveraging proven Methods

» We eat your dog food «



- **SAP Fiori** provides flexible generic UIs
- **OData** adds querying to REST
- **CDS** adds reflexive view building in addition

Best Practices & Benefits...

Capture Intent on App Domain Level

Accelerating Project Development

Leveraging proven Methods

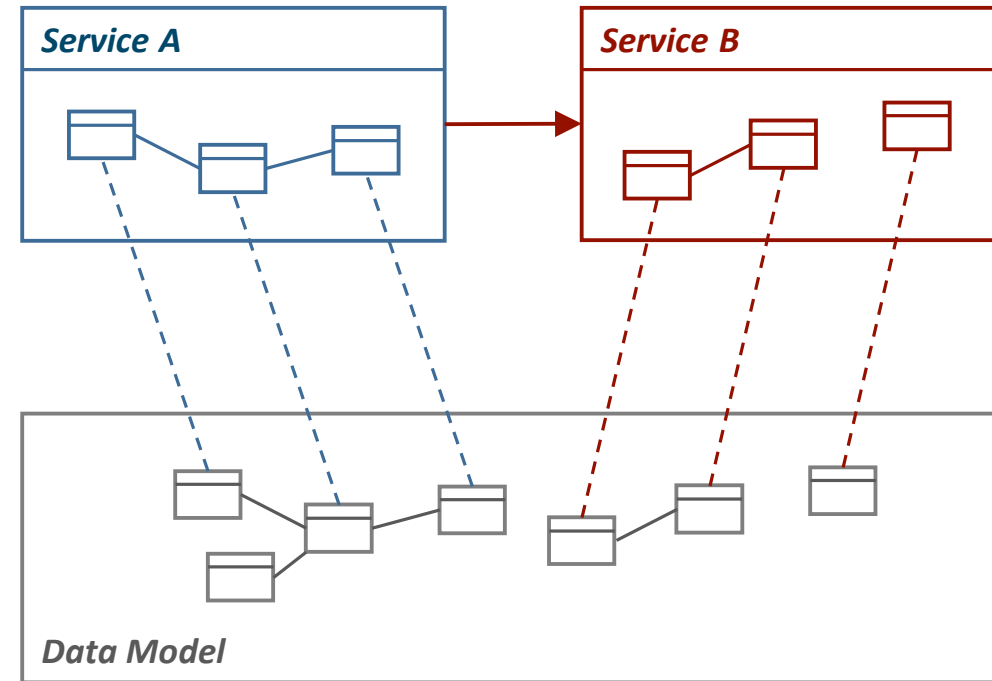
Service-oriented Paradigm

» *Services, not Business Objects* «

» *Queries, not ORM* «

→ Service Meshes

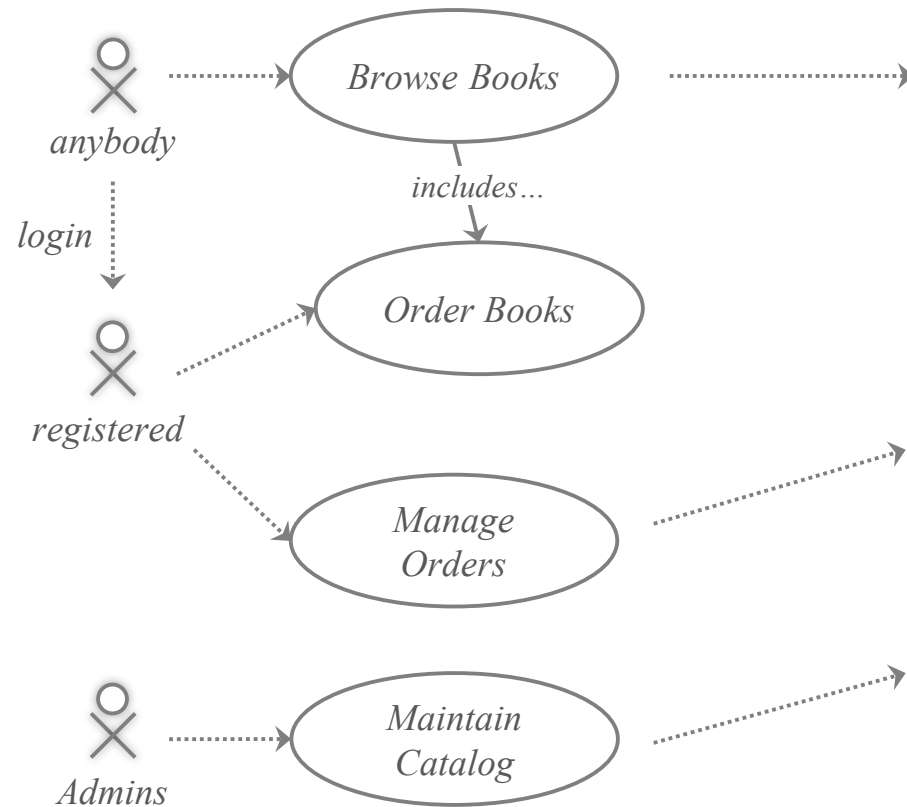
→ Late-cut Micro Services



Services are stateless API **facades**. They contain all logic and validations and frequently expose views on data.

Data Models define passive objects, mostly normalized and managed in databases

Single-purposed Services



```
service CatalogService {  
  expose entity Books @readonly;  
  expose entity Authors @readonly;  
  action order (book) @requires:$user;  
}
```

```
service CustomersService @(requires:$user) {  
  expose entity Orders @(restrict:[  
    {grant:'READ,DELETE', where:'buyer=$user'}  
  ]);  
}
```

```
service AdminService @(requires:'admin') {  
  expose entity Books;  
  expose entity Authors;  
  expose entity Orders;  
}
```

→ fosters focused and efficient implementations

→ enables 'late-cut μ services'



Best Practices & Benefits...

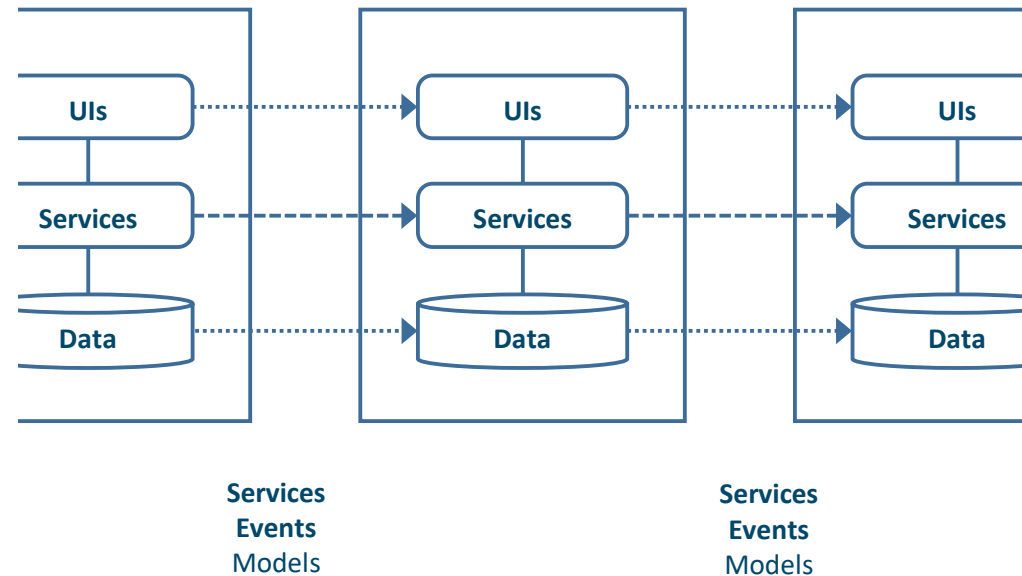
Capture Intent on App Domain Level

Accelerating Project Development

Leveraging proven Methods

Service-oriented Paradigm

Integration and Interoperability



» *Reuse, not re-invent* «

Best Practices & Benefits...

Capture Intent on App Domain Level

Accelerating Project Development

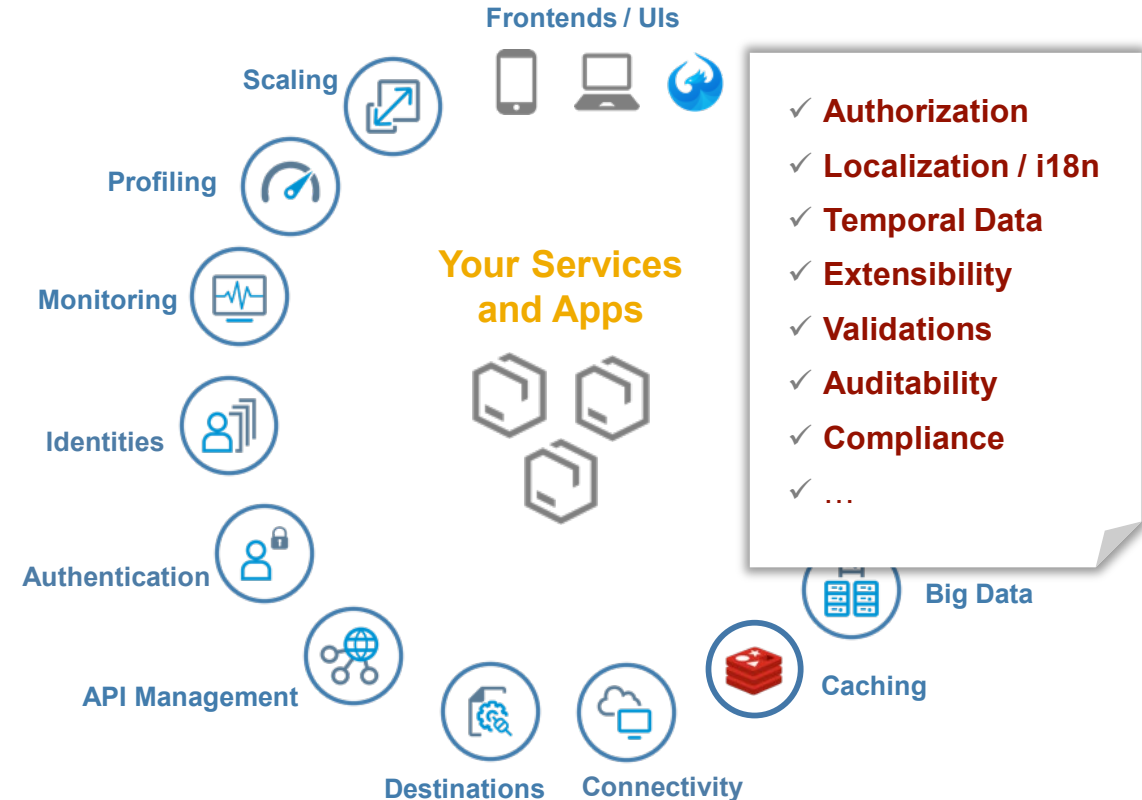
Leveraging proven Methods

Service-oriented Paradigm

Integration and Interoperability

Intrinsic Enterprise & Cloud Qualities

» *Focus on your domain* «



Best Practices & Benefits...

Capture Intent on App Domain Level

Accelerating Project Development

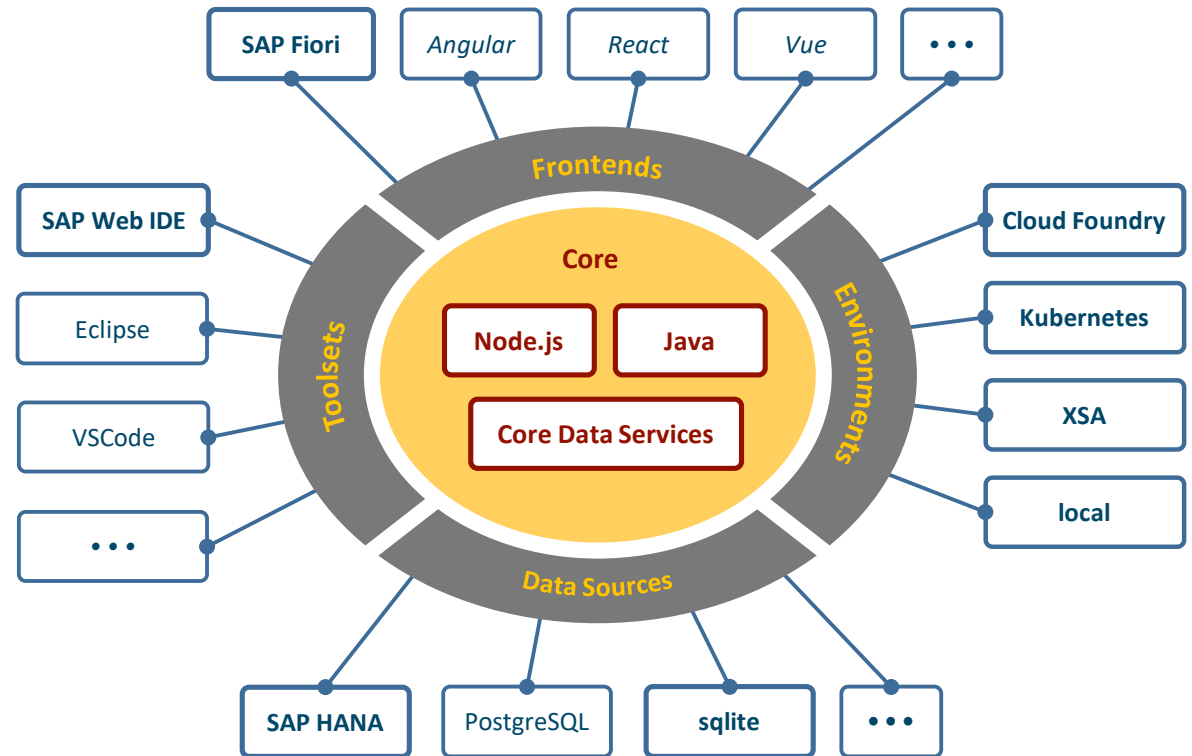
Leveraging proven Methods

Service-oriented Paradigm

Integration and Interoperability

Intrinsic Enterprise & Cloud Qualities

Preserving Openness & Flexibility



» *You stay in control; You choose!* «

Best Practices & Benefits...

Capture Intent on App Domain Level

Accelerating Project Development

Leveraging proven Methods

Service-oriented Paradigm

Integration and Interoperability

Intrinsic Enterprise & Cloud Qualities

Preserving Openness & Flexibility

SAP Web IDE Full-Stack

- *Project Storyboard*
- *Project Explorer*
- *Database Explorer*
- *Graphical Modeling*
- *Service Discovery*
- *Fiori Dev Tools*
- *CDS Editors*
- *Drag & Drop*



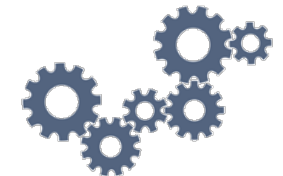
Local Development

- *Command-line Tools*
- *known IDEs / Editors*
- *w/ CDS Editors*
- *Local test runs*
- *Maven/npm-based automation*



Common Project Layout & Build

- *app*
- *srv*
- *db*



You can use Web IDE and/or local tools of your choice and switch between both.

Tailored tools with SAP Web IDE Next Generation*



- **Visual Studio Code-based**, open source, state of the art IDE
- **Containerized workspace**: With terminal access for full developer flexibility
- **CDS tools**: command line, CDS graphical modeler
- **Fiori tools**: Fiori template, UI5 development tools
- **SAP productivity tools**: Run Configuration, easy consumption of SAP services
- **Java & Node.js tools**: Full support for authoring, running, debugging

* beta coming soon

The screenshot displays the SAP Web IDE Next Generation interface. The top section shows a CDS model with a diagram of associations between entities. The bottom section features a terminal window with the following output:

```
> user@workspaces-ws-z8zn5-deployment-79669df5-6fbpb: ~/projects/bookshop x
/> successfully deployed database to ./my.db
user: bookshop $
user: bookshop $ cds run

[cds] - connect to datasource - sqlite:my.db
[cds] - serving CatalogService at /catalog
[cds] - service definitions loaded from:
```

Below the terminal, a table titled "Showing 5 'my_bookshop_Books' records" is displayed:

ID	title	stock	author_ID
201	Wuthering Heights	12	101
207	Jane Eyre	11	107
251	The Raven	333	150
252	Eleonora	555	150
271	Catweazle	22	170

Key Value Propositions

Guidance and Best Practices → opinionated and staying open and flexible

Intrinsic Enterprise & Cloud Qualities → taken care of out-of-the-box

Integration and Interoperability → easily reuse models and services

Accelerated Development → grow as you go & time to market

Focus on your Domain → what, not how; minimized boilerplate

Minimized Footprint → enabling serverless deployments

Safeguarded Investments → e.g. on journeys to new platform picks

Continue your SAP TechEd 2019 **Learning Experience**

Join the digital SAP TechEd Learning Room 2019 in [SAP Learning Hub](#)

- Access SAP TechEd **Learning Journeys**
- Discover **related** learning content
- Watch **webinars** of SAP TechEd lectures
- Learn about SAP's latest innovations with **openSAP**
- Collaborate with **SAP experts**
- **Self-test** your knowledge
- Earn a SAP TechEd **knowledge badge**

SAP Learning Hub

open**SAP**



Engage with the **SAP TechEd Community**

Access replays and continue your SAP TechEd discussion after the event within the **SAP Community**



Access replays

- Keynotes
- Live interviews
- Select lecture sessions

<http://sapteched.com/online>



Continue the conversation

- Read and reply to blog posts
- Ask questions
- Join discussions

sap.com/community



Check out the latest blogs

- See all SAP TechEd blog posts
- Learn from peers and experts

[SAP TechEd blog posts](#)

More information



Related SAP TechEd Learning Journeys

- CAA1 – Extend SAP solutions to support your intelligent enterprise
 - CAA2 – Build intelligent business solutions with SAP Cloud Platform
-

Related SAP TechEd sessions

- CAA106 – Walk-Through: SAP Cloud Application Programming Model on SAP Cloud Platform
 - CAA107 – Get to Know Core Data Services (CDS) in Depth
 - CAA160 – Getting Started with SAP Cloud Application Programming Model
 - CAA265 – Rapid Service Development with SAP Cloud Application Programming Model
 - CAA703 – Uncover Hidden Gems in SAP Cloud Application Programming Model
-

Public SAP Web sites

- Documentation: <https://cap.cloud.sap>
- SAP Community: <https://www.sap.com/community/topics/cloud-application-programming.html>



Thanks for attending **this session.**



Feedback

Please complete your session evaluation for **LT104**.

Contact for further topic inquiries

Daniel Hutzel
daniel.hutzel@sap.com

Christian Georgi
christian.georgi@sap.com

Follow us



www.sap.com/contactsap

© 2019 SAP SE or an SAP affiliate company. All rights reserved.

No part of this publication may be reproduced or transmitted in any form or for any purpose without the express permission of SAP SE or an SAP affiliate company.

The information contained herein may be changed without prior notice. Some software products marketed by SAP SE and its distributors contain proprietary software components of other software vendors. National product specifications may vary.

These materials are provided by SAP SE or an SAP affiliate company for informational purposes only, without representation or warranty of any kind, and SAP or its affiliated companies shall not be liable for errors or omissions with respect to the materials. The only warranties for SAP or SAP affiliate company products and services are those that are set forth in the express warranty statements accompanying such products and services, if any. Nothing herein should be construed as constituting an additional warranty.

In particular, SAP SE or its affiliated companies have no obligation to pursue any course of business outlined in this document or any related presentation, or to develop or release any functionality mentioned therein. This document, or any related presentation, and SAP SE's or its affiliated companies' strategy and possible future developments, products, and/or platforms, directions, and functionality are all subject to change and may be changed by SAP SE or its affiliated companies at any time for any reason without notice. The information in this document is not a commitment, promise, or legal obligation to deliver any material, code, or functionality. All forward-looking statements are subject to various risks and uncertainties that could cause actual results to differ materially from expectations. Readers are cautioned not to place undue reliance on these forward-looking statements, and they should not be relied upon in making purchasing decisions.

SAP and other SAP products and services mentioned herein as well as their respective logos are trademarks or registered trademarks of SAP SE (or an SAP affiliate company) in Germany and other countries. All other product and service names mentioned are the trademarks of their respective companies.

See www.sap.com/copyright for additional trademark information and notices.